

day-04-paperclip-closure

Day 04 - Paperclip 100% Closure

Date: 2026-05-10

Lane: Paperclip Workforce Control Plane

Status: SERVICE_DOWN closure with developer-side truth complete

Blocker class: SERVICE_DOWN

Blocker: 'paperclip_sandbox_service_not_running'

Phase 0-10 - Safety / Branch / Runtime

- Branch confirmed: 'to-knowledge-mc'.
- Runtime proof target: local production-style Mission Control runtime on '127.0.0.1:3337'.
- Mission Control runtime PID during proof: '15751'.
- No '.env' edits.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No Paperclip writes, task creation, mutations, or external execution attempted.
- Paperclip remains before OpenClaw+ in the operating chain.

Phase 10-20 - Inventory

Checked Paperclip-owned surfaces:

- 'GET /api/bridge/paperclip/status'
- 'GET /api/bridge/paperclip/companies'
- 'GET /api/bridge/paperclip/agents'
- 'GET /api/bridge/paperclip/issues'
- Paperclip bridge implementation and tests.
- Agent Network / Gateway Paperclip owner-facing status card.
- Local process list for Paperclip-like runtimes.
- Local service manager availability.
- Approved local workspace names for a runnable Paperclip repository.

Inventory result:

- No Paperclip process was running.
- No runnable Paperclip repo/package was found in the approved local workspace scan.
- Local 'systemctl' is unavailable in this macOS runtime context, so a user service cannot be started from here.
- The Mission Control Paperclip adapter expects a loopback or Tailnet-only service and refuses to connect on-local/non-Tailnet endpoints.

Phase 20-45 - Backend / Service Behavior

No source change was required for this phase because the backend already implements the required truthful behavior:

- Service detection checks only loopback/Tailnet-safe endpoints.
- Default probe target is loopback '3100'.
- Public endpoints are rejected.
- Unreachable service is classified as 'SERVICE_DOWN'.
- Company, agent, and issue reads return 503 when the sandbox service is not reachable.
- Protected actions remain disabled.
- Writes require Bridge Session and a configured Paperclip write adapter.

Current endpoint truth:

Route	HTTP	Result
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['/api/bridge/paperclip/status']	200	['SERVICE_DOWN', blocker 'paperclip_sandbox_service_not_running']
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'/api/bridge/paperclip/companies'	503	blocked by 'paperclip_sandbox_service_not_running'
'/api/bridge/paperclip/agents'	503	blocked by 'paperclip_sandbox_service_not_running'
'/api/bridge/paperclip/issues'	503	blocked by 'paperclip_sandbox_service_not_running'

Phase 45-65 - UI / Status Truth

Mission Control and Gateway behavior remains truthful:

- Paperclip is shown as Workforce Control Plane.
- Paperclip does not replace Agent Zero.
- Paperclip remains before OpenClaw+.
- Paperclip service health is not faked.
- Paperclip task creation remains blocked without Bridge Session and adapter proof.
- Owner-facing controls must remain live, Bridge-gated, or disabled with exact blocker.

No UI redesign was performed in this phase.

Phase 65-80 - Tests

Targeted Paperclip tests:

- 'src/lib/paperclip-bridge.test.ts'
- 'src/lib/paperclip-bridge-routes.test.ts'
- 'src/lib/paperclip-routing-gauntlet.test.ts'
- 'src/components/agent-network/AgentNetworkClient.paperclip.test.tsx'

Result:

- 4 test files passed.
- 47 tests passed.
- Paperclip routing gauntlet passed 1,000 scenarios with 0 failures.

Previously completed root validation during this closeout window:

- 'git diff --check' - PASS.
- 'pnpm run typecheck' - PASS.
- 'pnpm run build' - PASS.
- 'pnpm test' - PASS, 171 files / 1356 tests.
- Protected-file invariant scan - PASS.
- Owned-file secret scan - PASS.
- '.env' diff check - clean.
- Route rendering smoke - PASS, 46 routes and 8 designer pages checked.

Phase 80-95 - Runtime Proof

Authenticated Paperclip route proof used the Mission Control API key from the runtime database without printing it.

Observed:

- '/api/bridge/paperclip/status' is authenticated and reachable.
- The status endpoint reports the exact service blocker.
- Read routes for companies, agents, and issues do not fabricate rows when Paperclip is down.
- No raw private paths were exposed in the route summaries.
- No secrets were printed.
- No external writes occurred.

Phase 95-100 - Closeout Ledger

What was implemented:

- No new code was required. Day 04 closes developer-side as service-gated because Mission Control already exposes the correct Paperclip truth contract and test harness.

Files changed:

- 'runtime/day-04-paperclip-closure.md'
- 'runtime/day-04-paperclip-closure.pdf'

Routes/endpoints changed:

- None.

UI behavior:

- Paperclip remains visible but not falsely live.
- Paperclip controls remain blocked/gated according to the service and Bridge state.

Service/runtime behavior:

- Paperclip sandbox service is not running or reachable.
- No safe local/Tailnet-only Paperclip runtime could be started from this host context.
- No public exposure was added.

Remaining blocker:

- 'paperclip_sandbox_service_not_running'

Blocker classification:

- SERVICE_DOWN

Owner/admin action package:

1. Provide or start the Paperclip sandbox service on a loopback or Tailnet-only endpoint.
2. Expected health route: '/api/health'.
3. Expected read routes: '/api/companies', '/api/companies/{companyId}/agents', '/api/companies/{companyId}/issues'.
4. Do not expose Paperclip publicly.
5. Do not provide production secrets through '.env'.
6. After service is running, rerun the four Mission Control bridge routes listed above.

Rollback command:

- 'git revert <day-04-paperclip-report-commit>'

Next day:

- Day 05 OpenClaw+ starts automatically after this report commit and push.